CIVICSIFT AI

Named Entity Recognition (NER) Pipeline Design

WEEK 4

Project: CivicSift AI — Extracting useful information from citizen complaints

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1. Executive Summary

CivicSift AI is designed to analyse citizen complaints and extract important information in a structured form. This document proposes an end-to-end Named Entity Recognition (NER) pipeline for identifying people, places, organisations, dates, times, civic issues, and infrastructure references. The design supports multiple Indian languages and allows language-specific preprocessing. A multilingual transformer model is proposed for standard entities, while civic-specific labels can be added through fine-tuning, custom annotation, and carefully designed post-processing rules. The pipeline includes validation and human review so uncertain predictions are not treated as confirmed facts.

2. Introduction and Objectives

Named Entity Recognition is an NLP task that locates meaningful spans of text and assigns each span an entity category. For CivicSift AI, NER can convert an unstructured complaint such as "Water leakage near MG Road" into structured information that is easier to search, group, and analyse.

- Identify standard entities such as PERSON, LOCATION, ORGANIZATION, DATE, and TIME.
- Identify civic-specific entities such as ISSUE and INFRASTRUCTURE.
- Support complaints written in multiple Indian languages, including mixed-language text where feasible.
- Handle spelling variation, ambiguity, polysemy, and short complaint descriptions.
- Validate extracted entities and return structured output with confidence information.

3. Dataset Assessment and Data Strategy

The only dataset currently supplied for this project is nyc_311_complaints.csv. Its inspected structure is:

Property	Observed value
Rows	369
Columns	2 — text and complaint_type
Missing values	0
Unique complaint text values	348
Unique complaint types	107
Example complaint phrases	Pothole; Loud Music/Party; Blocked Hydrant; Blocked Sidewalk; Construction

Important limitation: this file contains short complaint phrases and record-level complaint categories. The complaint_type column is not token-level NER annotation. It cannot be directly treated as PERSON, LOCATION, DATE, or other token labels. The dataset also does not provide explicit language labels or full narrative complaints. Therefore, it is useful as a civic vocabulary and issue-category resource, but it is not sufficient by itself to train and evaluate a complete supervised NER model.

3.1 Recommended data strategy

1. Keep the supplied NYC 311 file as a source of civic terms, issue phrases, and category vocabulary.
2. Use a suitable public multilingual NER dataset for standard entity labels. Naamapadam (AI4Bharat) is a candidate to investigate; verify the exact release, languages, annotation schema, and licence before use.
3. Create a small manually annotated set of complete citizen-complaint sentences for CivicSift-specific ISSUE and INFRASTRUCTURE labels.
4. Split annotated data into training, validation, and held-out test sets. Keep test examples separate until final evaluation.
5. Use synthetic examples only to explain the design, not as real dataset evidence or evaluation results.

4. Entity Schema and Annotation

Entity label	Meaning	Illustrative example
PERSON	Name of a person	Rahul
LOCATION	Place, street, area, city, landmark	MG Road; Bengaluru
ORGANIZATION	Government body, department, company, institution	Municipal Corporation
DATE	Calendar date or date expression	10 November
TIME	Time expression	8:30 PM
ISSUE	Civic problem being reported	water leakage; pothole
INFRASTRUCTURE	Physical public asset or facility	road; hydrant; streetlight

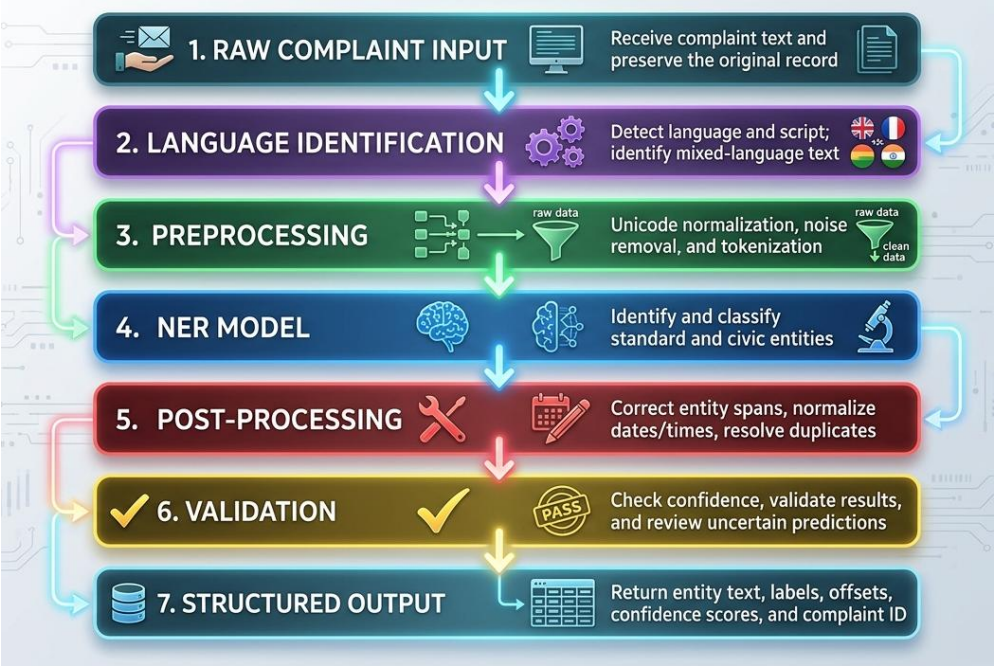
The standard labels are PERSON, LOCATION, ORGANIZATION, DATE, and TIME. ISSUE and INFRASTRUCTURE are custom CivicSift labels. The annotation guide must define consistent boundaries. For example, annotate “water leakage” as ISSUE and “pipeline” as INFRASTRUCTURE when both are present. If spans overlap, use a documented priority or allow nested spans only if the selected model and evaluation method support them.

4.1 BIO tagging

A common sequence-labelling format is BIO: B- means the beginning of an entity, I- means a continuation of that entity, and O means a token is outside all entities. For example, “Water leakage near MG Road” may be tagged as Water/B-ISSUE leakage/I-ISSUE near/O MG/B-LOCATION Road/I-LOCATION.

5. Proposed End-to-End Architecture

The following flow shows the proposed processing order:



5.1 Rationale for each component

Component	Why it is needed
Language identification	Selects suitable tokenization and model handling for multilingual input.
Preprocessing	Reduces formatting noise while preserving meaning and entity boundaries.
NER model	Learns contextual patterns to locate and classify entities.
Post-processing	Improves consistency, resolves common formatting issues, and normalizes values.
Validation and review	Detects invalid or uncertain predictions and supports quality improvement.
Structured output	Makes extracted information usable by downstream dashboards, search, and analytics.

6. Preprocessing Strategy

- Preserve the original complaint text so predictions can always be checked against the source.
- Apply Unicode normalization to handle equivalent character representations across Indian scripts.
- Remove irrelevant markup and repeated whitespace, but do not remove punctuation or numbers blindly because they may be part of addresses, dates, or identifiers.
- Detect language and script. For code-mixed complaints, retain the original tokens and avoid forcing the entire sentence into one language.
- Use language-appropriate tokenization. Keep token-to-character offsets so extracted spans can be mapped back to the original text.

- Handle spelling variants and transliteration with conservative normalization; do not automatically replace a word if the correction could change its meaning.
- Avoid removing stop words before transformer-based NER, because context words can help identify entity boundaries and meaning.

7. NER Model Selection and Training Strategy

A multilingual model is appropriate because the project expects complaints in multiple Indian languages. Candidate model families include mBERT (multilingual BERT), XLM-RoBERTa, and Indic-focused models such as IndicBERT. The final model should be selected by testing candidates on the same annotated validation set, rather than assuming one model is best.

Candidate	Reason to consider	Important check
mBERT	Multilingual pretrained transformer baseline.	Measure performance on the project's languages and short complaint text.
XLM-RoBERTa	Strong multilingual representation; suitable candidate for token classification.	Check compute requirements, language coverage, and validation results.
IndicBERT / Indic-focused model	Designed with Indian-language use cases in mind.	Verify supported languages, tokenizer behaviour, and availability of a suitable token-classification checkpoint.

7.1 Training approach

1. Prepare token-level annotations using the agreed entity schema and BIO format.
2. Align word-level labels with the model tokenizer's subword tokens; special tokens and continuation pieces must be handled consistently.
3. Fine-tune a token-classification model on the annotated training set. Use class weights or careful sampling only if imbalance is observed.
4. Tune hyperparameters and confidence thresholds on the validation set, not on the held-out test set.
5. Evaluate the final selected model once on the held-out test set and report results by entity type and language.

The custom ISSUE and INFRASTRUCTURE labels require suitable examples and annotations. A hybrid system can combine a multilingual transformer for standard entities with a custom fine-tuned component and/or carefully scoped gazetteer and pattern rules for civic vocabulary. Rules should support the model, not replace contextual analysis.

8. Post-processing and Entity Correction

- Repair invalid BIO sequences, such as an I-LOCATION token appearing without a preceding B-LOCATION, according to the annotation policy.

- Merge adjacent subword pieces belonging to the same predicted entity and recover the original text span.
- Normalize DATE and TIME expressions into a consistent format while retaining the original surface text.
- Use curated location names, civic asset terms, and department names as supporting gazetteers; do not assume a dictionary match is always an entity.
- Resolve duplicates and overlapping predictions using the documented span policy.
- Apply confidence thresholds. Send low-confidence or conflicting predictions to manual review instead of silently accepting them.

9. Contextual Ambiguity and Polysemy

Ambiguity occurs when a phrase can have more than one interpretation. Polysemy occurs when the same word has different related meanings depending on context. For example, “Park” may be a person's surname, a public park, or part of a place name. “Road” may refer to infrastructure, while “road blocked” describes an issue.

Challenge	Mitigation strategy
Same word, different meanings	Use surrounding words and sentence context; do not classify from a word list alone.
Short or incomplete complaints	Use complaint category and nearby terms as context, while marking uncertain cases.
Code-mixing and transliteration	Detect language/script, preserve original text, and include representative mixed-language examples in annotation.
Spelling variation	Use normalization and curated variants cautiously; retain original text.
Nested or overlapping spans	Define a clear annotation policy and apply it consistently during training and evaluation.
Rare entity labels	Collect more examples, monitor per-label recall, and use targeted annotation.

10. Error Handling and Accuracy Improvement

- Input validation: handle empty text, null values, unsupported encodings, and unusually long complaints safely.
- Language uncertainty: flag low-confidence language detection and route the record through a multilingual fallback where appropriate.
- Model failure: log the record ID and error type; do not fabricate entities when inference fails.
- Confidence review: send low-confidence predictions and conflicting spans to a human reviewer.
- Data quality: improve annotation guidelines, double-check difficult examples, and resolve disagreements.
- Model monitoring: inspect errors by language and entity label; add reviewed examples to later training cycles.

- Reproducibility: record model version, preprocessing version, dataset split, and evaluation configuration.

11. Validation and Evaluation

Evaluate NER at entity level, not merely by token accuracy. Exact-span precision measures how many predicted entities are correct; recall measures how many gold entities are found; F1 is the harmonic mean of precision and recall.

Precision = TP / (TP + FP) Recall = TP / (TP + FN) F1 = 2 × (Precision × Recall) / (Precision + Recall)

- Use exact entity-span matching with the correct label as the primary evaluation method.
- Report precision, recall, and F1 for each entity label and each supported language, along with overall micro/macro results where appropriate.
- Inspect false positives, false negatives, boundary errors, and confusion between ISSUE and INFRASTRUCTURE.
- Use a held-out test set that was not used for training or threshold tuning.
- Do not report accuracy or F1 scores until a properly annotated test set has actually been evaluated.

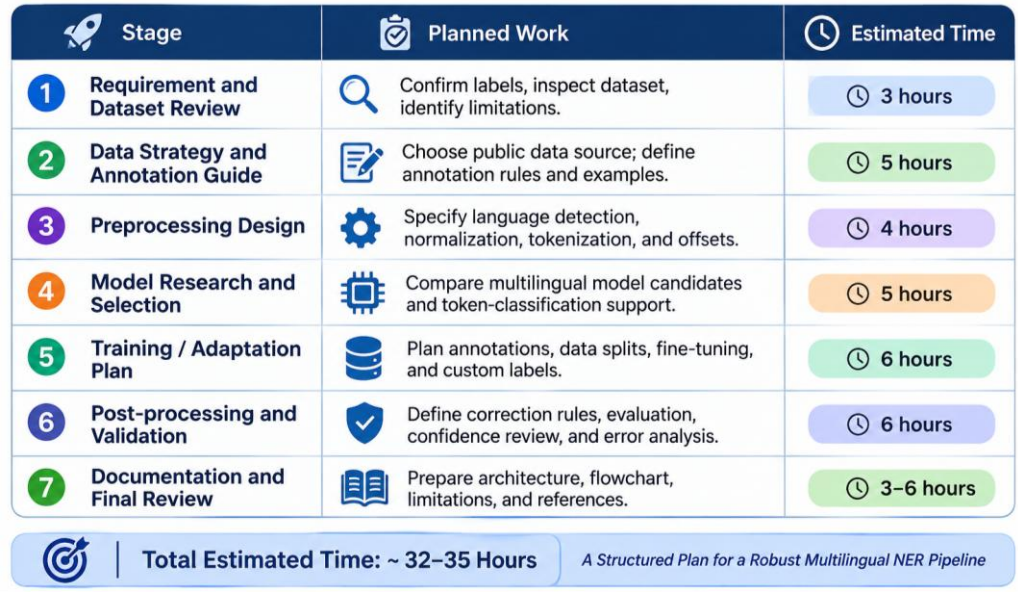
12. Example Structured Output

Illustrative example only — this is not a record from the supplied dataset and the confidence values below are placeholders, not measured model outputs.

```
{
 "complaint_id": "example_001",
 "language": "en",
 "entities": [
   {"text": "Water leakage", "label": "ISSUE"},
   {"text": "MG Road", "label": "LOCATION"},
   {"text": "Bengaluru", "label": "LOCATION"},
   {"text": "Rahul", "label": "PERSON"},
   {"text": "10 November", "label": "DATE"}
 ],
 "review_required": false
}
```

A production output should additionally include character offsets, confidence scores, model/version metadata, and the original complaint ID.

13. Implementation Roadmap (Estimated 30–35 Hours)



14. Conclusion

The proposed CivicSift AI NER pipeline combines multilingual text processing, contextual entity recognition, civic-specific labels, post-processing, and validation. The supplied NYC 311 dataset is useful for civic issue vocabulary but does not contain token-level NER annotations. A public multilingual NER resource and a manually annotated complaint sample are therefore recommended before model training and evaluation. This design provides a clear implementation path while avoiding unsupported claims about model accuracy.

15. References and Further Reading

- Devlin, J. et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. https://arxiv.org/abs/1810.04805
- Conneau, A. et al. (2020). Unsupervised Cross-lingual Representation Learning at Scale (XLM-R). https://arxiv.org/abs/1911.02116
- AI4Bharat Naamapadam dataset repository (verify release details and licence before use). https://github.com/AI4Bharat/naamapadam
- Hugging Face Transformers — Token classification task guide. https://huggingface.co/docs/transformers/tasks/token_classification
- spaCy — Linguistic features and named entities. https://spacy.io/usage/linguistic-features#named-entities
- NYC Open Data — 311 Service Requests. https://opendata.cityofnewyork.us/

Appendix A. Assumptions and Limitations

- The currently supplied file is nyc_311_complaints.csv with 369 rows and columns text and complaint_type.

- The file contains short complaint phrases and no token-level NER spans; complaint_type is not an NER tag.
- The file does not establish multilingual coverage. Multiple Indian languages are a project requirement, not a property verified in this file.
- Public dataset language coverage, annotation labels, and licence must be checked for the exact release selected.
- The document describes a pipeline design and implementation strategy. It does not claim that a model has been trained, deployed, or evaluated.